

GLASGOW BISHOPTON, United Kingdom

WMO: 031340

Lat: 55.9067N

Lon: 4.5325W

Elev: 59

StdP: 100.62

Time Zone: 0.00 (EUW)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.2	-2.6	-6.3	2.2	-2.0	-4.9	2.5	-0.9	14.4	7.1	12.8	8.1	1.6	330	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	22.9	17.3	21.1	16.2	19.5	15.4	18.1	21.5	17.1	19.9	16.2	18.6	3.7	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.7	12.0	19.3	15.9	11.4	18.3	15.1	10.8	17.3	51.1	21.6	48.1	20.0	45.5	18.6	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.6	9.4	8.4	DB	-5.9	26.6	2.1	2.1	-7.4	28.1	-8.6	29.4	-9.8	30.6	-11.3	32.1	(4)
(5)			WB	-6.3	20.0	2.1	1.0	-7.8	20.7	-9.0	21.3	-10.1	21.8	-11.6	22.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	4.1	4.4	5.7	8.2	10.7	13.5	15.3	14.5	12.7	9.6	6.3	4.4	(6)
(7)		DBStd	4.78	2.98	2.70	2.63	2.37	2.64	2.17	2.11	1.85	2.16	2.45	3.05	3.67	(7)
(8)		HDD10.0	889	183	157	135	63	22	1	0	0	4	36	114	175	(8)
(9)		HDD18.3	3359	442	390	392	303	237	144	99	119	170	270	361	433	(9)
(10)		CDD10.0	578	0	1	2	9	43	107	163	141	84	24	3	1	(10)
(11)		CDD18.3	6	0	0	0	0	0	1	4	1	0	0	0	0	(11)
(12)		CDH23.3	49	0	0	0	0	3	12	28	6	1	0	0	0	(12)
(13)		CDH26.7	4	0	0	0	0	0	2	2	1	0	0	0	0	(13)
(14)	Wind	WSAvg	3.8	4.3	4.1	4.3	4.1	3.8	3.5	3.3	3.5	3.6	3.6	3.8	3.9	(14)
(15)	Precipitation	PrecAvg	1170	130	93	102	64	70	65	68	88	117	129	125	124	(15)
(16)		PrecMax	1533	278	250	218	129	166	118	165	186	288	280	205	278	(16)
(17)		PrecMin	882	18	9	20	5	7	13	12	9	11	33	19	30	(17)
(18)		PrecStd	155	62	57	49	29	38	27	32	44	56	64	49	57	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.5	11.7	14.6	18.9	23.4	24.7	26.4	23.6	20.7	16.6	14.1	12.4	(19)
(20)			MCWB	10.4	10.0	10.2	13.0	16.6	17.8	19.2	18.4	16.0	14.3	12.9	11.4	(20)
(21)		2%	DB	10.2	10.4	11.9	16.0	20.1	21.8	23.4	20.9	18.6	15.2	12.7	11.3	(21)
(22)			MCWB	9.3	9.2	9.3	11.2	14.6	16.2	18.0	16.5	15.1	13.4	11.5	10.4	(22)
(23)		5%	DB	9.4	9.5	10.7	13.8	17.5	19.7	21.3	19.4	17.3	14.2	11.6	10.4	(23)
(24)			MCWB	8.5	8.3	8.6	10.0	12.9	15.0	16.7	15.7	14.5	12.7	10.5	9.6	(24)
(25)		10%	DB	8.6	8.7	9.7	12.4	15.4	17.9	19.5	18.1	16.2	13.3	10.7	9.5	(25)
(26)			MCWB	7.6	7.5	7.8	9.5	11.6	14.2	15.7	15.0	13.9	11.9	9.6	8.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	10.7	11.3	13.3	17.4	18.4	20.0	19.5	17.5	15.0	13.1	11.7	(27)
(28)			MCDWB	11.3	11.3	13.4	17.2	22.6	23.6	25.2	22.3	19.5	16.0	13.7	12.2	(28)
(29)		2%	WB	9.6	9.4	10.0	11.9	15.1	16.8	18.5	17.4	16.0	14.0	11.8	10.7	(29)
(30)			MCDWB	10.1	10.3	11.4	14.7	18.9	20.7	22.4	19.5	17.4	14.8	12.5	11.2	(30)
(31)		5%	WB	8.7	8.6	9.1	10.7	13.6	15.6	17.3	16.5	15.2	13.1	10.7	9.7	(31)
(32)			MCDWB	9.4	9.5	10.3	13.3	16.6	18.8	20.4	18.5	16.6	14.0	11.4	10.4	(32)
(33)		10%	WB	7.7	7.6	8.1	9.8	12.3	14.6	16.3	15.6	14.4	12.2	9.7	8.7	(33)
(34)			MCDWB	8.6	8.6	9.5	12.0	14.8	17.3	18.9	17.6	15.8	13.1	10.6	9.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.4	5.3	6.2	7.5	8.2	7.9	7.7	7.3	6.7	6.1	5.2	4.9	(35)
(36)			MCDBR	5.0	5.3	7.3	10.1	11.7	11.3	10.7	9.2	8.0	6.6	5.3	5.4	(36)
(37)			MCWBR	4.9	4.4	5.1	6.1	6.9	6.6	5.9	5.6	5.1	5.0	5.0	5.3	(37)
(38)		5% WB	MCDBR	5.0	4.9	6.3	8.9	10.3	10.1	9.5	8.1	6.6	5.7	5.4	5.4	(38)
(39)			MCWBR	5.0	4.5	5.0	5.8	6.5	6.3	5.6	5.5	4.8	4.9	5.3	5.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.330	0.342	0.366	0.383	0.386	0.382	0.383	0.367	0.359	0.342	0.321	0.315	(40)	
(41)		taud	2.208	2.312	2.289	2.281	2.318	2.363	2.365	2.420	2.445	2.422	2.300	2.184	(41)	
(42)		Ebn at Noon	551	717	791	834	854	862	852	844	799	713	568	468	(42)	
(43)		Edh at Noon	59	77	99	115	117	114	111	99	85	68	53	47	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.46	1.12	2.22	3.62	4.68	4.58	4.38	3.52	2.44	1.31	0.61	0.35	(44)	
(45)		RadStd	0.07	0.14	0.18	0.29	0.44	0.50	0.44	0.28	0.19	0.18	0.06	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page